5- In ______________________ chromosomes, the centromere is closer to one end than to the middle.
6- ______________________ is the term that describes a single set of chromosomes.
7- ______________________ is the process whereby both egg and pollen come from the same plant.
8- A phenomenon in which a single gene determines a number of distinct and seemingly unrelated characteristics is called ______________
9- ______________________ genes produce a subtle, secondary effect on phenotype by altering the effect of the alleles of other genes
10- A(n) ______________________ is a non-inherited change in phenotype arising from environmental effects that mimic the effect of a mutation in a gene.

List

A) Phenocopy B) Self fertilization, C) Permissive, D) Acrocentric,
E) monomorphic F) Haploid, G) Testcross, H) Penetrance,
I) Heterogeneous J) Pleiotropy, K) hypostatic, L) modifier
M) metacentric N) Diploid O) polygenic P) expressivity

1- O	2- G	3- H	4- W	5- M
6- A	7- B	8- I	9- P	10- C

III) True or false questions. Fill your answer in the table below (1 pt each)
1- Outbred population has identical alleles
2- Skin color is an example of a multifactorial inheritance
3- Redundant genes have similar biological function
4- Sickle cell anemia is a recessive lethal disease meaning that only heterozygous Hb^{S}/Hb^{A} die
5- O-blood individuals are universal donors

1- False	2- True	3- True	4- False	5- True

V) In sesame, the one-pod condition (P) is dominant to the three-pod condition (p), and normal leaf (L) is dominant over wrinkled leaf (l). The two characters are inherited independently. A cross between two members of the F1 generation produces the following progeny: 318 one-pod normal, 185 one-pod wrinkled, 323 three-pod normal and 184 three-pod wrinkled. Write the cross from F1 until F2, determine the phenotypic and the genotypic ratios. (4pts)

1) one : P three : p
2) normal leaf : L wrinkled leaf : l

PPLl PpLl Pp

normal : one : three : wrinkled
318 : 185 : 323 : 184

1 : 1 : 1 : 1

3

-14.5

Jordan University of Science and Technology
Department of Applied Biological Science
First Exam., Mol. Genetics B341)
Summer Semester, 2016-2017

Serial#

Student Name: [redacted]
Student Number: [redacted]

I) Multiple choice questions. (1 pt/question). Choose the correct answer for the questions below and fill your answer in the table

1- In a monohybrid cross AA X aa, what proportion of homozygotes is expected among the F 2 offspring?

AA x aa
Aa x Aa
AA, Aa

A. 1/4
B. 1/2
C. 3/4
D. All are homozygotes.
E. None are homozygotes.

2- An interaction between non-allelic genes that results in the masking of expression of a phenotype is
A. epistasis.
B. epigenetics.
C. dominance.
D. codominance.
E. incomplete dominance.

3- Which of the following phenotypic ratios show independent assortment?
A. 9:3:3:1
B. 9:7
C. 9:3:4
D. 13:3
E. All of the choices are correct.

4- Dextral shell female snail with (dd) genotype mated to Sinistral shell male with (DD) genotype results in
A. all offspring dextral
B. all offspring sinistral
B. 3 sinistral to 1dextral
D. 1 sinistral to 3 dextral
E. none of the above

... of mating cross?

A) Phenocopy
E) monomorphic
I) Heterogenous
M) metacentric
B) Self fertilization,
F) Haploid,
J) Pleiotropy,
N) Diploid
C) Permissive,
G) Testcross,
K) hypostatic,
O) polygenic
D) Acrocentric,
H) Penetrance,
L) modifier
P) expressivity

4.5

1- O	2- G	3- H	4- ~~K~~	5- M
6- A	7- B	8- I	9- P	10- C

III) True or false questions. Fill your answer in the table below (1 pt/each)

1- Outbred population has identical alleles
2- Skin color is an example of a multifactorial inheritance
3- Redundant genes have similar biological function
4- Sickle cell anemia is a recessive lethal disease meaning that only heterozygoutes $Hb\beta^S/Hb\beta^A$ die
5- O-blood individuals are universal donors

1- False	2- True	3- True	4- False	5- True

V) In sesame, the one-pod condition (P) is dominant to the three-pod condition (p), and normal leaf (L) is dominant over wrinkled leaf (l). The two characters are inherited independently. A cross between two members of the F1 generation produces the following progeny: 318 one-pod normal, 185 one-pod wrinkled, 323 three-pod normal and 184 three-pod wrinkled. Write the cross from F1 until F2, determine the phenotypic and the genotypic ratios. (4pts)

one : P three : p

5- In ____________________ chromosomes, the centromere is closer to one end than to the middle.
6- ____________________ is the term that describes a single set of chromosomes.
7- ____________________ is the process whereby both egg and pollen come from the same plant
8- A phenomenon in which a single gene determines a number of distinct and seemingly unrelated characteristics is called ____________________.
9- ____________________ genes produce a subtle, secondary effect on phenotype by altering the effect of the alleles of other genes
10- A(n) ____________________ is a non-inherited change in phenotype arising from environment effects that mimic the effect of a mutation in a gene.

List

A)Phenocopy	B) ~~Self fertilization,~~	C) Permissive,	D) Acrocentric,
E) monomorphic	F) Haploid,	G) Testcross,	H) Penetrance,
I) Heterogenous	J) Pleiotropy,	K) hypostatic,	L) modifier
M)metacentric	N)Diploid	O)polygenic	P)expressivity

1- O	2- G	3- H	4- K	5- M
6- A	7- B	8- J	9- P	10- C

18/25

excuse me if maybe 18

Exam time: 60 minutes
4/3/2019

Jordan University of Science and Technology
Department of Applied Biological Science
First Exam. Mol. Genetics B341
Second Semester, 2018/2019

Serial# 19

Student Name:-- Sarah bilal Malkawi
Student Number:-- 118907

I) Multiple choice questions. (1 pt/question). Choose the correct answer for the questions below, then fill your answer in the table below

1) A parental plant is allowed to self-pollinate (self-fertilization). All of the offspring in the F1 and F2 are identical in phenotype to the parental plant. What is the genotype of the parental plant?
a) heterozygous b) homozygous dominant c) hybrids
d) homozygous recessive e) true breeding

2) The random distribution of alleles to the gametes that occurs when genes are located on different chromosomes is best described by the:
a) Principle of segregation b) principle of independent assortment
c) law of haploid gamete d) theory of reproduction
e) principle of segregation and principle of independent assortment are correct

B>b T>t

3) Black fur (B) in mice is dominant to brown fur (b). Short tails (T) is dominant to long (t) tails. What proportion of the offspring from a cross between an individual with the genotype BbTt and BBtt will have black fur and long tails? B-tt
a) 1/16 b) 3/16 c) 6/16 d) 8/16 e) 9/16

$P(BBtt) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$P(Bbtt) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$\frac{2}{4} = \frac{1}{2}$

4) Why the human genome project is considered a milestone in genetics?
a) Because it revealed how human's genome is different than the genome of other organisms
b) Because it revealed how different organisms have very similar genome size
c) Because it revealed how different organisms have very similar DNA sequences
d) Because it confirmed that human was evolved from chimpanzee
e) Both b and c are correct

5) Which of the following is not a reason why analyzing human genetics is more difficult than analyzing plant genetics?
a) genetic analysis of human traits relies on family records
b) it is not ethical to set up controlled human matings to observe progeny
c) humans produce more progeny than plants
d) humans produce many fewer progeny than plants
e) some genetic conditions do not manifest themselves until middle age or beyond

6) A heritable factor that exists in two forms is also known as a(n):
a) Allele b) Gene c) Factor d) Organism e) Heterozygote

7) As a result of the ``Human Genome project``, the genome/s of which organism/s was/were determined?
a) Human b) *D. melanogaster* c) *E. coli* d) *M. muscalis*
e) all of the above

8) Neonatal chondrodysplasia punctata (CDP) is characterized by epiphyseal stippling and midfacia hypoplasia. CDP is usually inherited, but can be acquired because of maternal vitamin K deficiency. Th phenotypic pattern can be described as

a) Incomplete penetrance
b) Complete penetrance
c) phenocopy
d) Variable expressivity
e) polygenic

9) Red snapdragons are crossed with white snapdragons and in the F1 generation this cross produces pink offspring. The allele (W) controlling the red color is considered to be:
a) Recessive
b) Dominant
c) Incompletely dominant
d) Codominant
e) None of these

10) Among 10 individuals with the same HD mutation only 6 of them show Huntington disease phenotype. Based on the above information, which of the following statements is/are correct?
a) Huntington disease has variable expressivity
b) The HD mutation is 60% penetrant
c) Huntington disease is pleotropic
d) Huntington disease is polygenic
e) all of the above are correct

Name

Sarah

6

ZIPGRADE.COM

1 A B C D E
2 A B C D E
3 A B C D E
4 A B C D E
5 A B C D E
6 A B C D E
7 A B C D E
8 A B C D E
9 A B C D E
10 A B C D E

Student ID

1	9
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

First_winter_2018_2019 (2541)

8
5

II) Fill in the blank with one of the suggested list below. (1 pt/each)

1-A gene with more than one allele is called (E) polymorphic

2- Environmental conditions that prevent conditional lethals to live are known as (D) Restrictive

3- A phenomenon in which a single gene determines a number of distinct and seemingly unrelated characteristics is called (J) Pleiotropy

4- epistatic (M) gene is masked by the effect of another gene

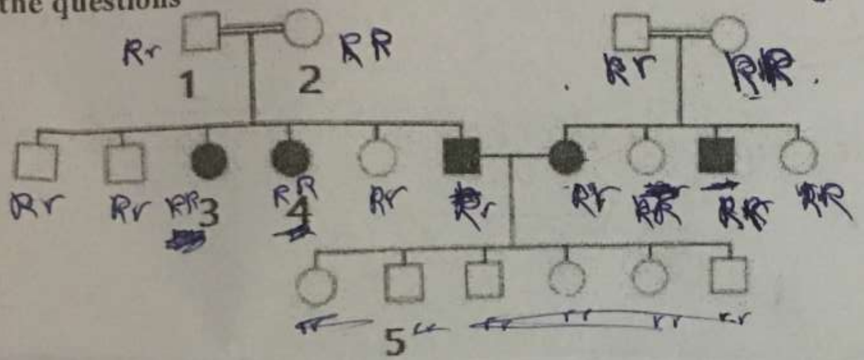
5- multifactorial (N) phenotypes that is controlled by both genetic and environmental factors.

List

A)Phenocopy	~~B) Self fertilization~~	~~C) Permissive~~	D) Restrictive
~~E)~~ polymorphic	~~F) Haploid~~	G) Testcross,	H) Penetrance
I) Heterogenous	J) Pleiotropy	~~K) hypostatic~~	L) modifier
M)epistatic	N)multifactorial	~~O)polygenic~~	P)expressivity

more same gene

II) Study the pedigrees below constructed from a family suffering from a rare disease, then answer the questions



RR Rr

Not reccessip

RR Rr rr
1 : 2 : 1

1) What is the probable type (mode) of inheritance for this trait? (1pt) Redundancy

~~recessive~~ ~~Dominant need 2 alleles to show disease~~ ~~Redundancy~~

2) What is the genotype of individual II-3, III-5?(0.5pt/each)

II-3 : ~~Rr~~ RR   III-5 : rr

3) What is the probability that III-5 is not a carrier, individual I-1 is a carrir? (0.5pt/each)

0   1/2   100%   1/2

III) Assay questions

1) Mention three reasons why individuals with the same mutation may not show the same/identical expected phenotype? (1.5 pts)

1) Depending on the biochemical process it contributes to

2) ~~one gene controls many phenotypes~~ 2) different environmental conditions

3) modifier

4) Penetrance & expressivity

1

$w > y$

2) Two true-breeding white and yellow strains of plants were mated, and the F1 progeny were all white. When the F1 plants were allowed to self-fertilize, 302 white-flowered, 76 green-flowered and 24 yellow flowered F2 plants grew.

Dominant

12 : 3 : 1

$wwyy$

A. How could you describe the inheritance of flower color? (1pt) Dominant epistasis

B. Describe how specific alleles influence each other and therefore affect phenotype. (1pt)

C. You picked a green flowered plant from the F2 progeny. Explain how you can identify the genotype of this plant? (write down all possibilities). (2pts)

if one Dominant (white) allele is present it mask the other 2 alleles of trait.

colorless precursor $\xrightarrow{\text{Enzyme G } GG/Gy}$ Green ⟵ Enzyme W WW/Ww doesn't allow ~~deposition~~ Green to deposite — white 302

$ww \rightarrow$ allows deposition of Green pigment.

~~colorless pre~~ Enzym gg ~~Enzyme~~ X green

~~Cont~~ complementary testing.

green flowered $yy\ W\_$

3) Three alleles at gene I determine a person's ABO blood group phenotype (I^A, I^B and i). A couple have 3 children. The mother has type AB blood and the father has type A blood. Blood tests on the 3 childrens reveal that the first child has type B blood, the second child has type A blood and the third appears to have type O blood.

There is no question of paternity or maternity (i.e the children are indeed the children of this couple).

$I^A I^A$ or $I^A i$

A) What are the possible ABO genotypes of the father? Child 2? (1)

B) What is the likely reason for child 3 apparently having type O blood? (1) → because it has hh ↓ doesn't have the substance that allows I^A or I^B to combine wi

C) What is the term to describe the dominance relationships between I^B and I^A? (1)

A) father

co-dominance. if $I^A I^B$ Poly morphic

♀ AB $I^A I^B\ Hh$

♂ A $I^A I^A$ | $I^A i$ Hh

I^A | I^A i

I^A I^B

B | 2 A $I^A I^A HH$

appers O hh

I

3

A) Father
$I^A I^A\ Hh$
$I^A i\ Hh$

Child 2
$I^A I^A\ HH$
$I^A I^A\ Hh$
$I^A i\ HH$
$I^A i\ Hh$

A. 9:3:3:1 B. 9:7 C. 9:3:4

D. 13:3 E. All of the choices are correct.

4- Dextral shell female snail with (dd) genotype mated to Sinistral shell male with (DD) genotype results in

A. all offspring dextral B. all offspring sinistral

B. 3 sinistral to 1dextral D. 1 sinistral to 3 dextral E. none of the above

dd DD

5) Which of the following terms is not a type of mating cross?

A) test-cross B) reciprocal C) monohybrid D) dominant E) dihybrid

6) The alleles of blood groups A, B and O show

A. complete dominance. B. recessiveness. C. codominance.

D. complete dominance, recessiveness, and codominance E. none of the choices is correct.

dd

7) The Human Genome project was responsible for sequencing the genome(s) of

A. Humans. B. *D. melanogaster* C. *E. coli* D. *M. muscalis*

E. all of the above

8) Which molecule acts as an intermediary in the processing of the genetic information?

A. DNA B. RNA C. Fat D. Protein E. phosphate groups

4

1- B	2- ~~A~~ B	3- A	4- A	5- D
6- D	7- A	8- B	9-	10-

II) Fill in the blank with one of the suggested list below. Fill your answer in the table blow. (0.5 pt/each)

1-A gene with only one wild-type allele is called________.

2- A mutation in any one of a number of genes that can give rise to the same phenotype is termed a(n)________________ trait.

3- Environmental conditions that allow conditional lethals to live are known as ________

4- If all individuals carrying a dominant mutant gene show the mutant phenotype, the gene is said to show complete ________________.

145

10/23

Jordan University of Science and Technology
Department of Applied Biological Science
First Exam., Mol. Genetics B341)
Summer Semester, 2016-2017

Serial#

Student Name: [redacted]
Student Number: [redacted]

I) Multiple choice questions. (1 pt/question). Choose the correct answer for the questions below and fill your answer in the table

1- In a monohybrid cross AA X aa, what proportion of homozygotes is expected among the F 2 offspring?
A. ¼ B. ½ C. ¾
D. All are homozygotes. E. None are homozygotes.

AA - aa
Aa - Aa
AA, Aa, Aa

2- An interaction between non-allelic genes that results in the masking of expression of a phenotype is
A. epistasis. B. epigenetics. C. dominance.
D. codominance. E. incomplete dominance.

3- Which of the following phenotypic ratios show independent assortment?
A. 9:3:3:1 B. 9:7 C. 9:3:4
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A. all offspring dextral B. all offspring sinistral
B. 3 sinistral to 1dextral D. 1 sinistral to 3 dextral E. none of the above

5) Which of the following terms is not a type of mating cross?
A) test-cross B) reciprocal C) monohybrid D) dominant E)dihybrid

6) The alleles of blood groups A, B and O show
A. complete dominance. B. recessiveness. C. codominance.
D. complete dominance, recessiveness, and codominance E. none of the choices is correct.

7) The Human Genome project was responsible for sequencing the genome(s) of
A. Humans. B. *D. melanogaster* C. *E. coli* D. *M. muscalis*
E. all of the above

8) Which molecule acts as an intermediary in the processing of the genetic information?
A. DNA B. RNA C. Fat D. Protein E. phosphate groups

1- B	2- ~~A~~ B	3- A	4- A	5- D
6- D	7- A	8- B	9-	10-

II) Fill in the blank with one of the suggested list below. Fill your answer in the table blow. (0.5 pt/each)
1-A gene with only one wild-type allele is called________.
2- A mutation in any one of a number of genes that can give rise to the same phenotype is termed a(n)________________ trait.
3- Environmental conditions that allow conditional lethals to live are known as ____________
4- If all individuals carrying a dominant mutant gene show the mutant phenotype, the gene is said to show complete ________________.

dd - DD
Dd